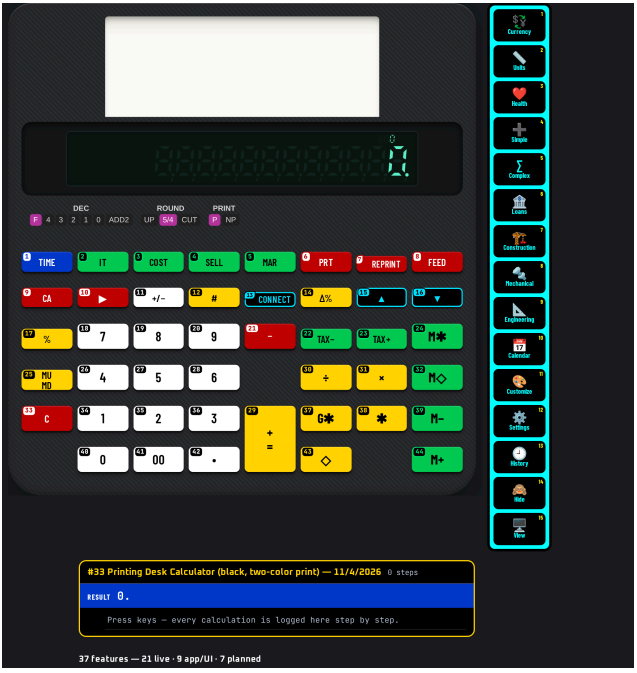
U-45 CALCULATORS

User Manual

Calculator #33 Printing Desk Calculator (black, two-color print)

Desk – printing



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#33 — Printing Desk Calculator (black, two-color print)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 TIME]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

adding-machine logic (+= and −= post amounts to the tape; ◇/T totals).

Chapters

1. Getting started
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1 · Getting started

Open U-45 CALCULATORS, choose **#33** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#11, #18, #19, #20, #26, #27, #28, #34, #35, #36, #40, #41, #42
op	Operators & execute	#12, #14, #17, #25, #29, #30, #31, #37, #38, #43
fn	Functions	#1
mem	Memory, variables & registers	#2, #3, #4, #5, #22, #23, #24, #32, #39, #44
clr	Clear, delete & power	#6, #7, #8, #9, #10, #21, #33
mod	Shift, modes, menus & navigation	#13, #15, #16

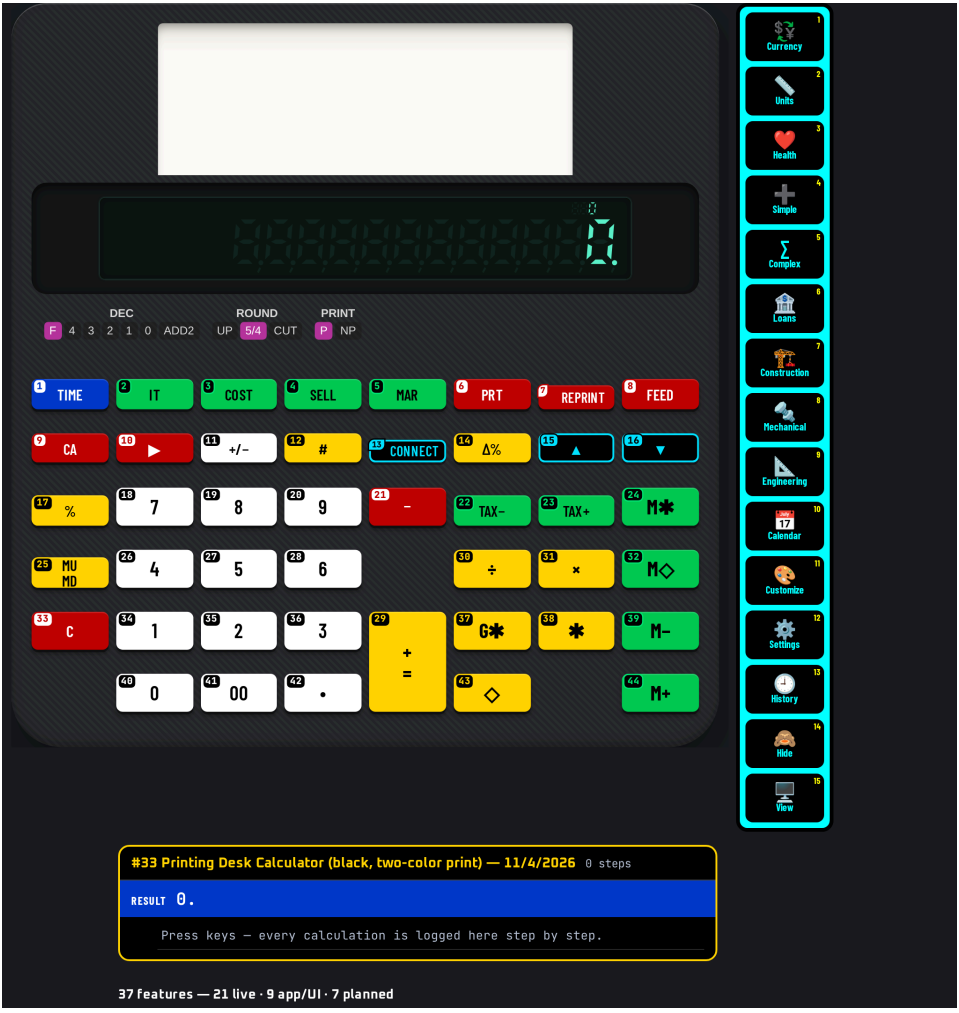
2 · Features

37 features: 21 live on the keys, 9 app/UI features, 7 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **On-screen paper tape with negative values in red**
2. LIVE **Adding-machine logic +=, −=, ✱ total, ◇ subtotal**
3. LIVE **Item count IT**
4. LIVE **Memory M+ M− M◇ M✱**
5. LIVE **Grand total G✱**
6. LIVE **MU / MD mark-up / mark-down**
7. LIVE **COST / SELL / MAR margin**
8. LIVE **TAX+ / TAX− (rate via ▲ ▼, default 8 %)**
9. LIVE **Δ% change**
10. LIVE **Decimal slide F 4 3 2 1 0 ADD2**
11. LIVE **Rounding slide UP / 5/4 / CUT**
12. LIVE **Non-add # / date print**
13. LIVE **REPRINT and FEED**
14. LIVE **TIME / DATE from system clock**
15. LIVE **00 / 000 fast-zero keys**
16. LIVE **Percent key (x %, add-on and discount chains)**
17. LIVE **Percent change Δ% between two values**
18. LIVE **Sign change (−) / +/− / CHS**
19. LIVE **Independent memory M+**
20. LIVE **Independent memory M−**

21. LIVE Backspace / DEL editing of the current entry
22. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
23. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
24. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
25. APP/UI Show / hide key numbers with one button (remembered between sessions)
26. APP/UI Copy the current display text to the clipboard
27. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
28. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
29. APP/UI Reference photo viewer and tool-tip name on every key
30. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
31. PLANNED Currency conversion
32. PLANNED Tape export to PDF / CSV
33. PLANNED Tape search / filter
34. PLANNED Tape save & reload
35. PLANNED Speech read-out
36. PLANNED Colour-blind theme
37. PLANNED Custom tax presets

3 · Numbered key map



Calculator #33 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	TIME	·	·	·	fn
2	IT	·	·	·	mem
3	COST	·	·	·	mem
4	SELL	·	·	·	mem
5	MAR	·	·	·	mem
6	PRT	·	·	·	clr
7	REPRINT	·	·	·	clr
8	FEED	·	·	·	clr
9	CA	·	·	·	clr
10	▶	·	·	·	clr
11	+/−	·	·	·	num
12	#	·	·	·	op
13	CONNECT	·	·	·	mod
14	Δ%	·	·	·	op
15	▲	·	·	·	mod
16	▼	·	·	·	mod
17	%	·	·	·	op
18	7	·	·	·	num
19	8	·	·	·	num
20	9	·	·	·	num
21	−	·	·	·	clr
22	TAX−	·	·	·	mem
23	TAX+	·	·	·	mem

#	Face	Shift	Alpha	3rd	Group
24	M✱	·	·	·	mem
25	MUMD	·	·	·	op
26	4	·	·	·	num
27	5	·	·	·	num
28	6	·	·	·	num
29	+=	·	·	·	op
30	÷	·	·	·	op
31	×	·	·	·	op
32	M◇	·	·	·	mem
33	C	·	·	·	clr
34	1	·	·	·	num
35	2	·	·	·	num
36	3	·	·	·	num
37	G✱	·	·	·	op
38	✱	·	·	·	op
39	M−	·	·	·	mem
40	0	·	·	·	num
41	00	·	·	·	num
42	•	·	·	·	num
43	◇	·	·	·	op
44	M+	·	·	·	mem

4 · Key-by-key reference

Every key on calculator #33, in number order, with every function it carries in this app.

#1 TIME `Functions`

Primary: TIME — print the time

#2 IT `Memory, variables & registers`

Primary: IT — item count

#3 COST `Memory, variables & registers`

Primary: COST

#4 SELL `Memory, variables & registers`

Primary: SELL

#5 MAR `Memory, variables & registers`

Primary: MAR — margin %

#6 PRT `Clear, delete & power`

Primary: PRT — print the display on the tape

#7 REPRINT `Clear, delete & power`

Primary: REPRINT — reprint the last calculation

#8 FEED `Clear, delete & power`

Primary: FEED — paper feed (blank line)

#9 CA `Clear, delete & power`

Primary: CA — clear all (including accumulator)

Keyboard: `Escape`

#10 ▶ `Clear, delete & power`

Primary: Backspace — delete the last digit typed

Keyboard: `Backspace` `Delete`

#11 +/− `Digits & number entry`

Primary: Change sign (+/−) of the number being typed or the current result

#12 # `Operators & execute`

Primary: # — non-add (print a reference number)

#13 CONNECT `Shift, modes, menus & navigation`

Primary: Printed label only (no action in this app)

#14 Δ% Operators & execute

Primary: Δ% — percent change

#15 ▲ Shift, modes, menus & navigation

Primary: Tax rate ▲ — increase the tax rate setting

Keyboard: ArrowUp

#16 ▼ Shift, modes, menus & navigation

Primary: Tax rate ▼ — decrease the tax rate setting

Keyboard: ArrowDown

#17 % Operators & execute

Primary: % — percent (with + / − gives add-on / discount)

Keyboard: %

#18 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#19 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#20 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#21 – Clear, delete & power

Primary: Subtraction –

Keyboard: -

#22 TAX– Memory, variables & registers

Primary: TAX– — remove tax

#23 TAX+ Memory, variables & registers

Primary: TAX+ — add tax at the set rate

#24 M✱ Memory, variables & registers

Primary: M✱ — memory total (recall and clear)

#25 MUMD Operators & execute

Primary: MU — mark-up / mark-down

#26 **4** `Digits & number entry`

Primary: Digit 4

Keyboard: [4]

#27 **5** `Digits & number entry`

Primary: Digit 5

Keyboard: [5]

#28 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: [6]

#29 **+=** `Operators & execute`

Primary: += — add the entry to the accumulator (adding-machine logic)

Keyboard: [+] [Enter] [=]

#30 **÷** `Operators & execute`

Primary: Division ÷

Keyboard: [/]

#31 **×** `Operators & execute`

Primary: Multiplication ×

Keyboard: [*] [x]

#32 M◇ Memory, variables & registers

Primary: M◇ — memory subtotal (recall)

#33 C Clear, delete & power

Primary: C — clear the entry

#34 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#35 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#36 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#37 G✱ Operators & execute

Primary: GT — grand total

#38 ✱ Operators & execute

Primary: ✱ — total (print and clear)

#39 M− Memory, variables & registers

Primary: M− — subtract the value from memory M

#40 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#41 00 Digits & number entry

Primary: Double zero — types “00” in one press (fast entry of amounts)

#42 • Digits & number entry

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: . ,

#43 ◇ Operators & execute

Primary: ◇ — subtotal (print, keep running)

#44 M+ `Memory, variables & registers`

Primary: M+ — add the value to independent memory M

5 · Worked examples

11 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

#19 8 → #40 0 → #31 × → #34 1 → #27 5 → #17 %

Display shows 12. **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.2 — Adding a list (tape)

Calculate 125 + 48 − 19, total. Press Reset first.

#34 1 → #35 2 → #27 5 → #29 + = → #26 4 → #19 8 → #29 + = → #34 1 → #20 9 → #21 − → #38 ✱

Display shows 154. **— expected 154. ✔ verified on the app.**

Each amount is posted with its sign; the total key prints and clears the running total.

Example 5.3 — Multiplying on an adding machine

Calculate 12 × 15. Press Reset first.

#34 1 → #35 2 → #31 × → #34 1 → #27 5 → #29 + =

Display shows 180. **— expected 180. ✔ verified on the app.**

Multiplication is completed with the +=/= key, as on desk adding machines.

Example 5.4 — Division on an adding machine

Calculate 144 ÷ 12. Press Reset first.

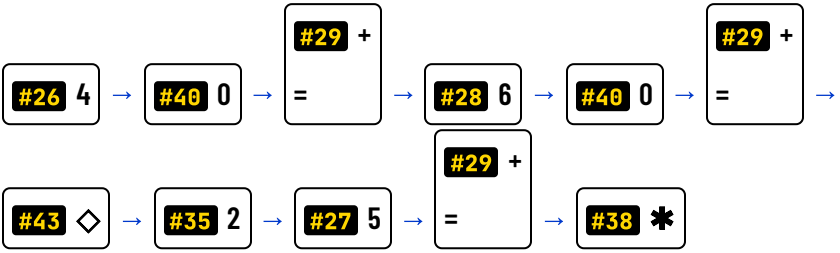
#34 1 → #26 4 → #26 4 → #30 ÷ → #34 1 → #35 2 → #29 + =

Display shows 12. **— expected 12. ✔ verified on the app.**

Division is completed with the +=/= key.

Example 5.5 — Subtotal then continue

Calculate 40 + 60 (subtotal), + 25, total. Press Reset first.

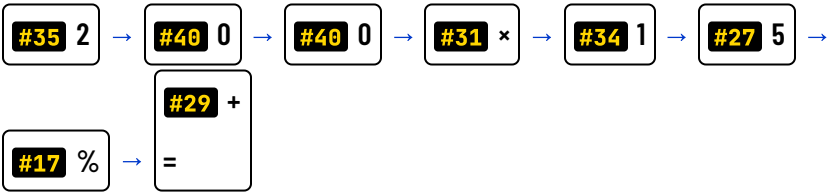


Display shows 125. **— expected 125. ✔ verified on the app.**

The subtotal key prints the running total without clearing it; the total key prints and clears.

Example 5.6 — Add a percentage (price + %)

Calculate 200 + 15 %. Press Reset first.



Display shows 230. **— expected 230. ✔ verified on the app.**

× then % works out the percentage part; the + key then adds it to the base (price plus percentage).

Example 5.7 — Take off a percentage (discount)

Calculate **200 − 15 %**. Press **Reset** first.

#35 2 → #40 0 → #40 0 → #31 × → #34 1 → #27 5 → #17 % → #21 −

Display shows `170.` — expected 170. ✔ verified on the app.

× then % works out the percentage part; the − key then takes it off the base.

Example 5.8 — What percentage is one number of another

Calculate **30 as a % of 120**. Press **Reset** first.

#36 3 → #40 0 → #30 ÷ → #34 1 → #35 2 → #40 0 → #17 %

Display shows `25.` — expected 25. ✔ verified on the app.

÷ then % gives the first number as a percentage of the second.

Example 5.9 — Percentage change (Δ%)

Calculate **change from 80 to 100**. Press **Reset** first.

#19 8 → #40 0 → #14 Δ% → #34 1 → #40 0 → #40 0 → #14 Δ%

Display shows `25.` — expected 25. ✔ verified on the app.

Δ% is pressed after the old value and again after the new value; a fall shows as a negative percentage.

Example 5.10 — Margin from cost and selling price

Calculate cost 60, sell 80 → margin %. Press Reset first.

#28 6 → #40 0 → #3 COST → #19 8 → #40 0 → #4 SELL

Display shows 25. **— expected 25. ✔ verified on the app.**

Enter cost with COST and selling price with SELL — the margin is worked out as profit ÷ selling price.

Example 5.11 — Invoice lines in memory

Calculate 3 × 12 + 5 × 8 (memory total). Press Reset first.

#36 3 → #31 × → #34 1 → #35 2 → #44 M+ → #27 5 → #31 × → #19 8 → #44 M+ → #24 M✱

Display shows 76. **— expected 76. ✔ verified on the app.**

M+ finishes each line and adds it to memory; M✱ (memory total) recalls the sum and clears memory.

6 · Tricks & hacks

Hack 1: Adding machine logic

Type an amount then +=; type the next amount then += … finally ✱ prints the total.

Hack 2: Subtotal check

◇ prints a running subtotal without clearing.

Hack 3: Fast zeros

Use 00 / 000 for round amounts.

Hack 4: Tax

Set the rate once with the rate keys; then TAX+ adds it to any amount.

Hack 5: Mark-up

COST, SELL, MAR solve for the missing one of cost, selling price and margin.

Hack 6: Decimal slide

ADD2 places the decimal point for you (type cents only).

Hack 7: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #44.

Hack 8: Type faster on a computer

20 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 9: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 10: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 11: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
Escape	#9	CA	CA — clear all (including accumulator)
Backspace Delete	#10	▶	Backspace — delete the last digit typed
ArrowUp	#15	▲	Tax rate ▲ — increase the tax rate setting
ArrowDown	#16	▼	Tax rate ▼ — decrease the tax rate setting
%	#17	%	% — percent (with + / − gives add-on / discount)
7	#18	7	Digit 7
8	#19	8	Digit 8
9	#20	9	Digit 9
-	#21	−	Subtraction −
4	#26	4	Digit 4
5	#27	5	Digit 5
6	#28	6	Digit 6
+ Enter =	#29	+=	+= — add the entry to the accumulator (adding-machine logic)
/	#30	÷	Division ÷
* x	#31	×	Multiplication ×
1	#34	1	Digit 1
2	#35	2	Digit 2
3	#36	3	Digit 3
0	#40	0	Digit 0
. ,	#42	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Pressing = for totals

Adding machines use ✱ for total and ◇ for subtotal; += adds entries.

Mistake 2: Decimal slide setting

If 1234 shows as 12.34, the ADD2 slide is on.

Mistake 3: Wrong tax rate

Check the RATE annunciator; the default rate may not be yours.

Mistake 4: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 5: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (96)

Work each one on calculator #33, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 8 % of 250 = _____
2. 25 % of 360 = _____
3. 12.5 % of 40 = _____
4. 10 % of 450 = _____
5. 20 % of 125 = _____
6. 5 % of 640 = _____
7. 75 % of 48 = _____
8. 30 % of 90 = _____
9. 1200 + 350 − 75, total = _____
10. 45.5 + 12.25 − 7.75, total = _____
11. 600 + 240 − 85, total = _____
12. 18 + 27 − 36, total = _____
13. 2500 + 1250 − 625, total = _____
14. 3.5 + 4.25 − 1.75, total = _____
15. 77 + 88 − 99, total = _____
16. 1000 + 1 − 1, total = _____

17. 480 + 120 − 360, total = _____

18. 64 + 36 − 50, total = _____

19. 12.75 + 8.4 − 3.15, total = _____

20. 305 + 207 − 109, total = _____

21. 1500 + 2500 − 400, total = _____

22. 42 + 58 − 25, total = _____

23. 730 + 270 − 999, total = _____

24. 55.55 + 44.45 − 10, total = _____

25. 7 × 125 = _____

26. 24 × 3.5 = _____

27. 36 × 18 = _____

28. 250 × 4 = _____

29. 9.5 × 6 = _____

30. 48 × 25 = _____

31. 11 × 11 = _____

32. 16 × 16 = _____

33. 3.2 × 5 = _____

34. 45 × 12 = _____

35. 0.75 × 80 = _____

36. 99 × 9 = _____

37. 125 × 8 = _____

38. 60 × 24 = _____

39. 13 × 17 = _____

40. 1000 ÷ 8 = _____

41. 625 ÷ 25 = _____

42. 99 ÷ 4 = _____

43. 1500 ÷ 6 = _____

44. 360 ÷ 12 = _____

45. 84 ÷ 7 = _____

46. 1 ÷ 8 = _____

47. 2400 ÷ 15 = _____

48. 91 ÷ 13 = _____

49. 450 ÷ 9 = _____

50. 125 + 75 (subtotal), + 300, total = _____

51. 9.5 + 0.5 (subtotal), + 5, total = _____

52. 1000 + 500 (subtotal), + 250, total = _____

53. 80 + 25 % = _____

54. 1250 + 8 % = _____

55. 45 + 20 % = _____

56. 64 + 50 % = _____

57. 320 + 12.5 % = _____

58. 150 + 6 % = _____

59. 80 − 25 % = _____

60. 1250 − 8 % = _____

61. 45 − 20 % = _____

62. 64 − 50 % = _____

63. 320 − 12.5 % = _____

64. 150 − 6 % = _____

65. 45 as a % of 180 = _____

66. 7 as a % of 28 = _____

67. 150 as a % of 600 = _____

68. 9 as a % of 12 = _____

69. 66 as a % of 88 = _____

70. change from 40 to 50 = _____

71. change from 1200 to 1500 = _____

72. change from 15 to 18 = _____

73. change from 500 to 625 = _____

74. cost 75, sell 100 → margin % = _____

75. cost 120, sell 200 → margin % = _____

76. cost 36, sell 48 → margin % = _____

77. cost 400, sell 500 → margin % = _____

78. cost 9, sell 12 → margin % = _____

79. cost 150, sell 250 → margin % = _____

80. cost 70, sell 140 → margin % = _____

81. 2 × 45 + 4 × 7.5 (memory total) = _____

82. 10 × 3.2 + 6 × 1.5 (memory total) = _____

83. 12 × 25 + 3 × 40 (memory total) = _____

84. 7 × 7 + 8 × 8 (memory total) = _____

85. 5 × 99 + 2 × 49.5 (memory total) = _____

86. 15 % of 80 = _____

87. 125 + 48 − 19, total = _____

88. 12 × 15 = _____

89. 144 ÷ 12 = _____

90. 40 + 60 (subtotal), + 25, total = _____

91. 200 + 15 % = _____

92. 200 − 15 % = _____

93. 30 as a % of 120 = _____

94. change from 80 to 100 = _____

95. cost 60, sell 80 → margin % = _____

96. 3 × 12 + 5 × 8 (memory total) = _____

Set B — Key-finding drills (33)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “TIME” (TIME)? _____
2. Which key is “IT” (IT)? _____
3. Which key is “COST” (COST)? _____
4. Which key is “SELL” (SELL)? _____
5. Which key is “MAR” (MAR)? _____
6. Which key is “PRT” (PRT)? _____
7. Which key is “REPRINT” (REPRINT)? _____
8. Which key is “FEED” (FEED)? _____
9. Which key is “CA” (CA)? _____
10. Which key is “▶” (Backspace)? _____
11. Which key is “+/−” (Change sign)? _____
12. Which key is “#” (#)? _____
13. Which key is “Δ%” (Δ%)? _____
14. Which key is “ ▲ ” (Tax rate ▲)? _____
15. Which key is “ ▼ ” (Tax rate ▼)? _____
16. Which key is “%” (%)? _____
17. Which key is “−” (Subtraction −)? _____
18. Which key is “TAX−” (TAX−)? _____

19. **Which key is “TAX+” (TAX+)?** _____

20. **Which key is “M✱” (M✱)?** _____

21. **Which key is “MUMD” (MU)?** _____

22. **Which key is “+=” (+=)?** _____

23. **Which key is “÷” (Division ÷)?** _____

24. **Which key is “×” (Multiplication ×)?** _____

25. **Which key is “M◇” (M◇)?** _____

26. **Which key is “C” (C)?** _____

27. **Which key is “G✱” (GT)?** _____

28. **Which key is “✱” (✱)?** _____

29. **Which key is “M−” (M−)?** _____

30. **Which key is “00” (Double zero)?** _____

31. **Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)?** _____

32. **Which key is “◇” (◇)?** _____

33. **Which key is “M+” (M+)?** _____

Appendix A — Drill answers

Set A

1. **20** (display `20.`)

   `#35` 2 → `#27` 5 → `#40` 0 → `#31` × → `#19` 8 → `#17` %

2. **90** (display `90.`)

   `#36` 3 → `#28` 6 → `#40` 0 → `#31` × → `#35` 2 → `#27` 5 → `#17` %

3. **5** (display `5.`)

   `#26` 4 → `#40` 0 → `#31` × → `#34` 1 → `#35` 2 → `#42` • → `#27` 5 → `#17` %

4. **45** (display `45.`)

   `#26` 4 → `#27` 5 → `#40` 0 → `#31` × → `#34` 1 → `#40` 0 → `#17` %

5. **25** (display `25.`)

   `#34` 1 → `#35` 2 → `#27` 5 → `#31` × → `#35` 2 → `#40` 0 → `#17` %

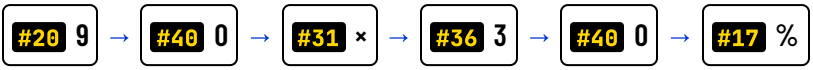
6. **32** (display `32.`)

   `#28` 6 → `#26` 4 → `#40` 0 → `#31` × → `#27` 5 → `#17` %

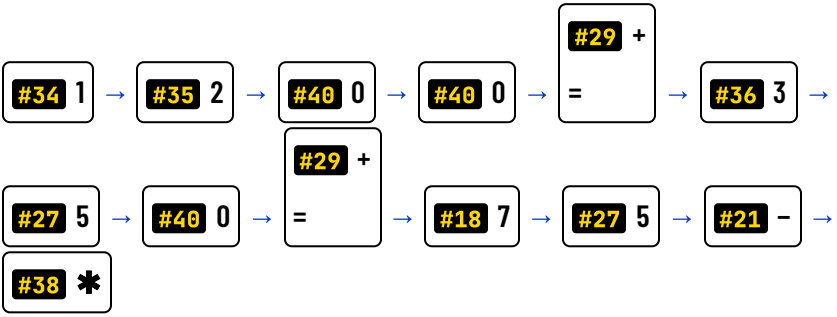
7. **36** (display `36.`)

   `#26` 4 → `#19` 8 → `#31` × → `#18` 7 → `#27` 5 → `#17` %

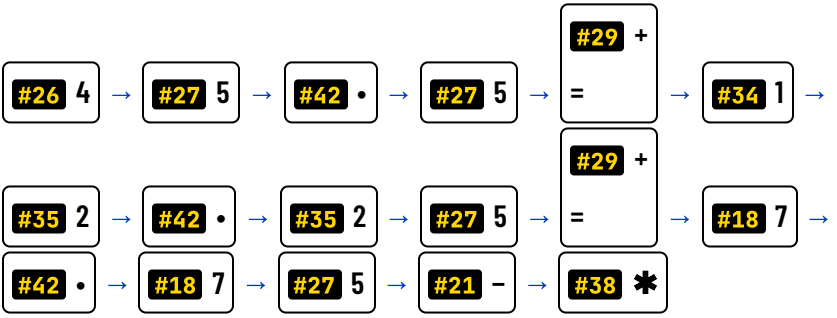
8. 27 (display `27.`)



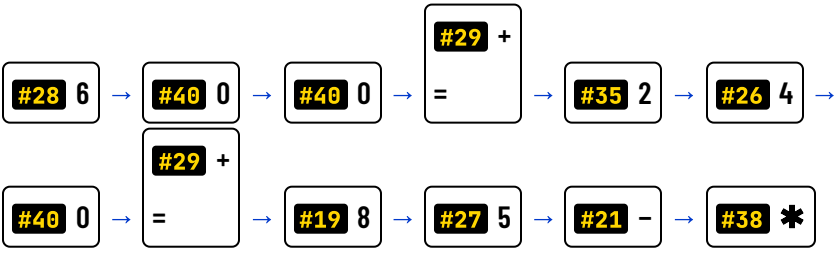
9. 1475 (display `1,475.`)



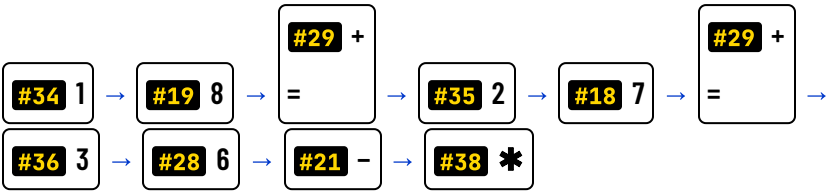
10. 50 (display `50.`)



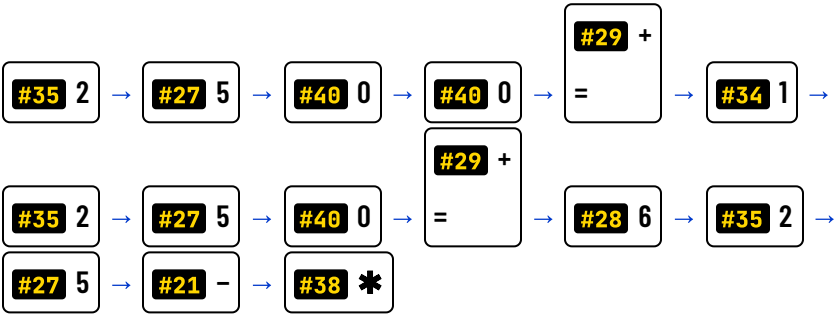
11. 755 (display `755.`)



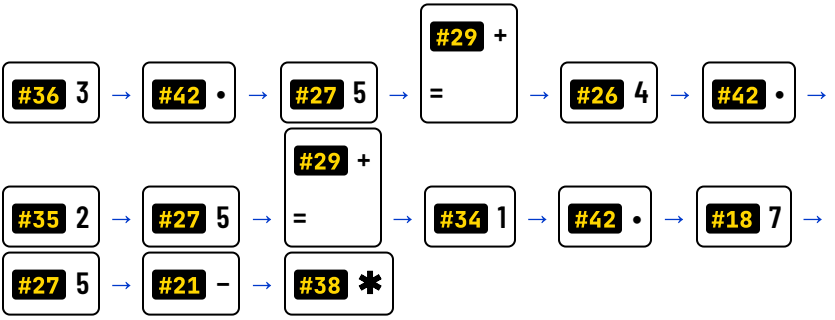
12. 9 (display 9.)



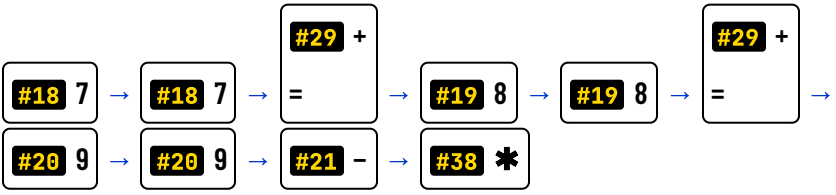
13. 3125 (display 3,125.)



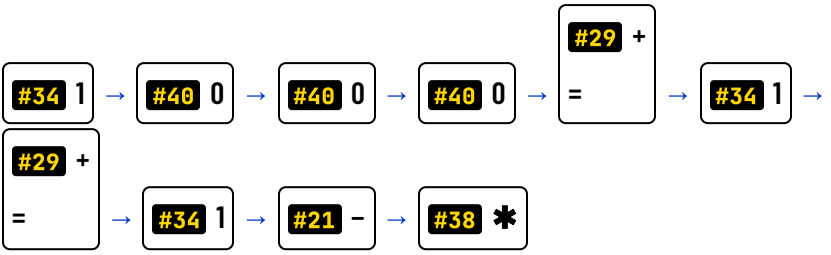
14. 6 (display 6.)



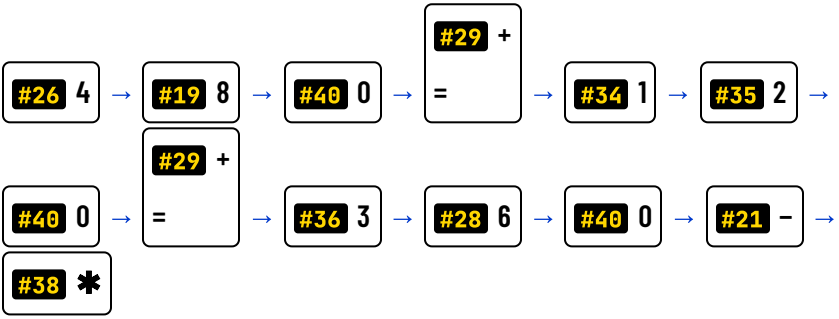
15. 66 (display 66.)



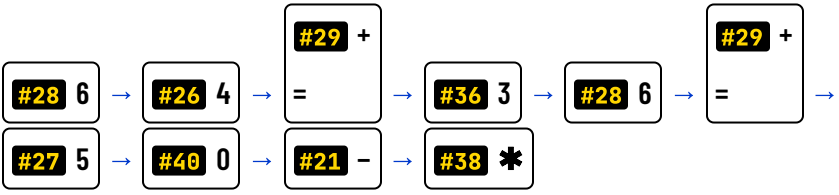
16. 1000 (display 1,000.)



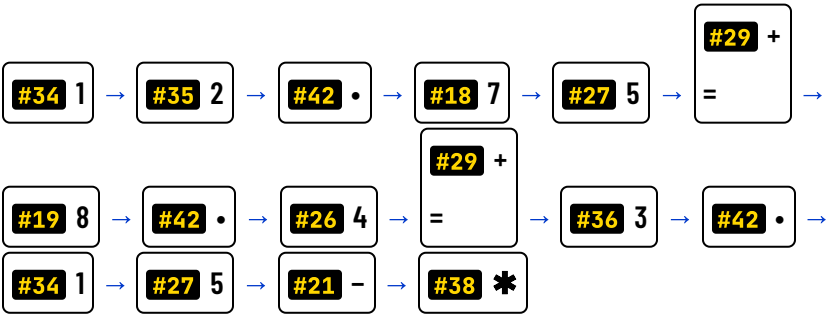
17. 240 (display 240.)



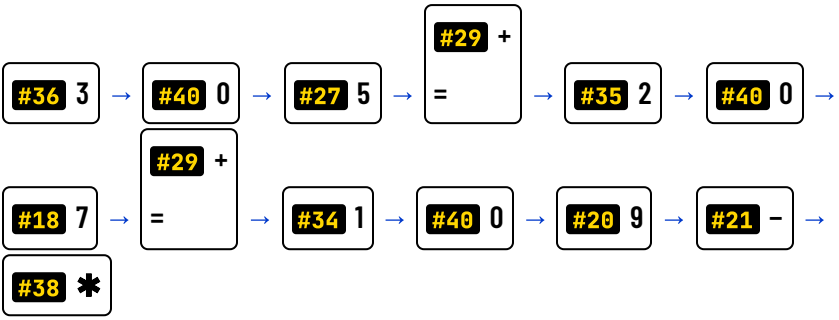
18. 50 (display 50.)



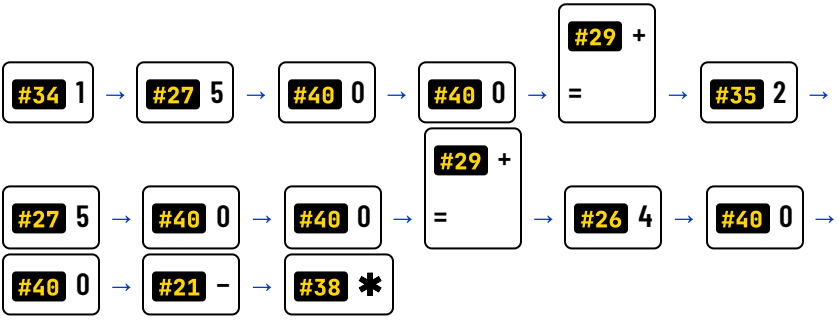
19. 18 (display 18.)



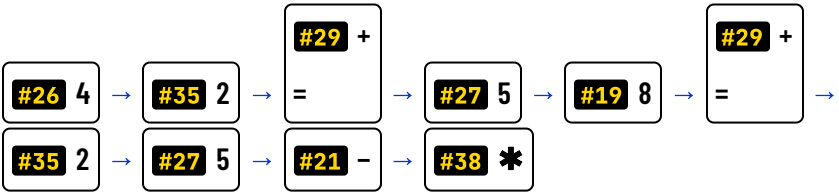
20. 403 (display 403.)



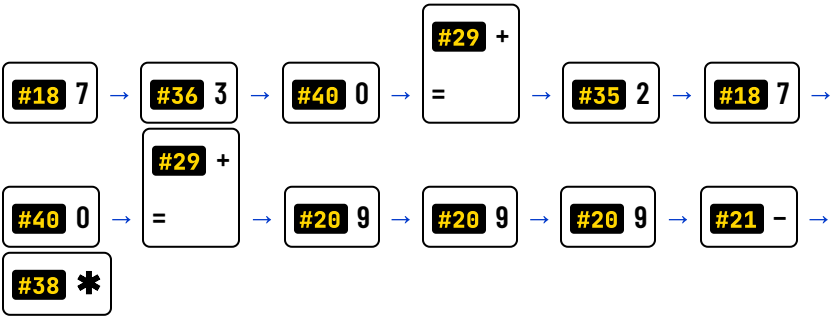
21. 3600 (display 3,600.)



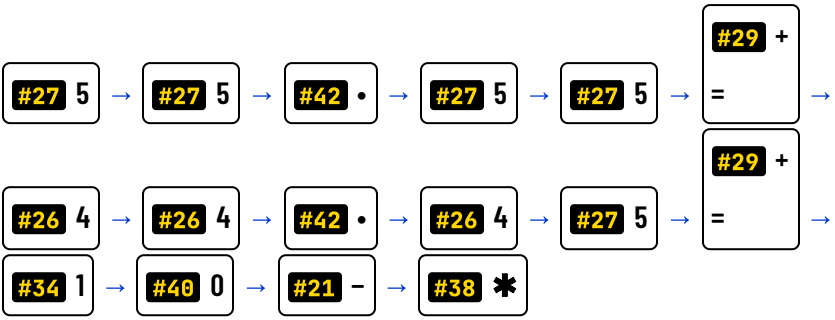
22. 75 (display 75.)



23. 1 (display 1.)



24. 90 (display 90.)



25. 875 (display 875.)



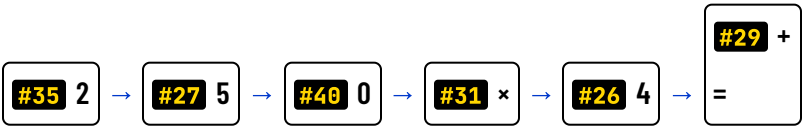
26. 84 (display 84.)



27. 648 (display 648.)



28. 1000 (display 1,000.)



29. 57 (display `57.`)



30. 1200 (display `1,200.`)



31. 121 (display `121.`)



32. 256 (display `256.`)



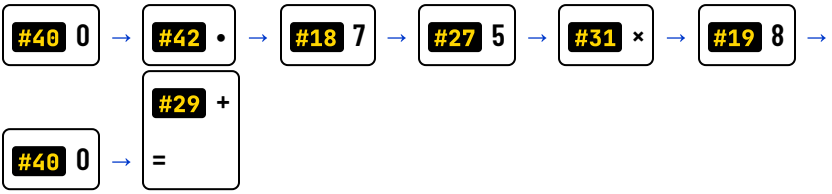
33. 16 (display `16.`)



34. 540 (display `540.`)



35. 60 (display `60.`)



36. 891 (display `891.`)



37. 1000 (display `1,000.`)



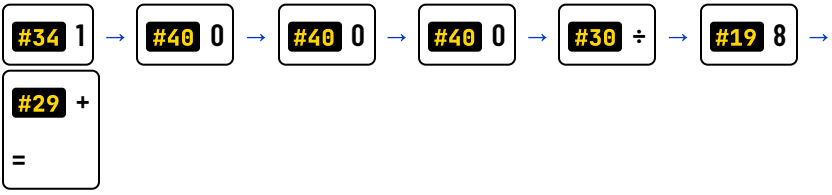
38. 1440 (display `1,440.`)



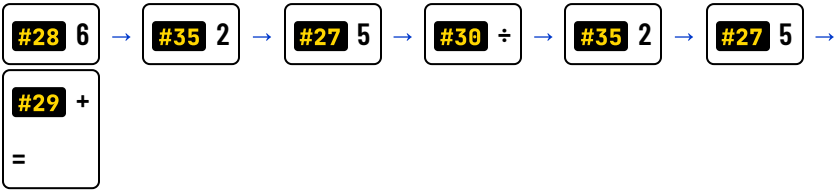
39. 221 (display `221.`)



40. 125 (display `125.`)



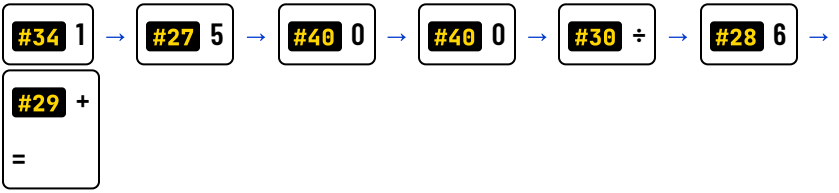
41. 25 (display `25.`)



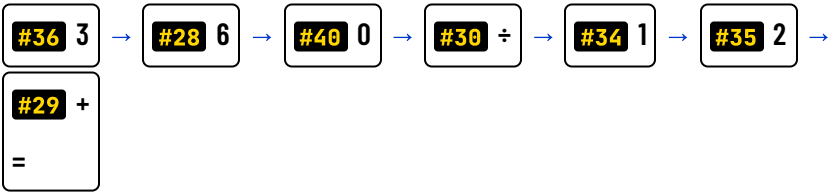
42. 24.75 (display `24.75`)



43. 250 (display `250.`)



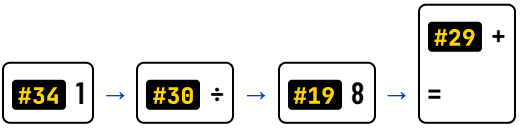
44. 30 (display `30.`)



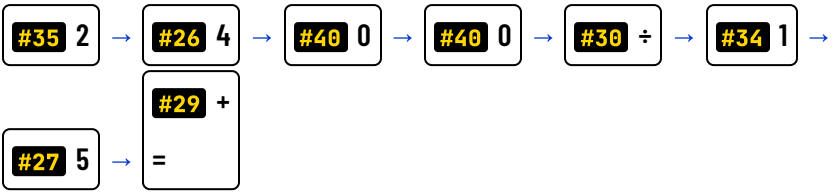
45. 12 (display `12.`)



46. 0.125 (display `0.125`)



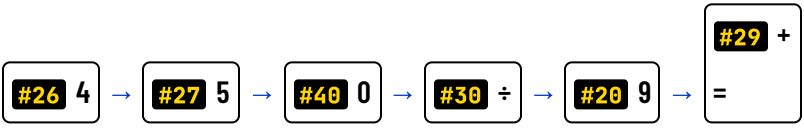
47. 160 (display 160.)



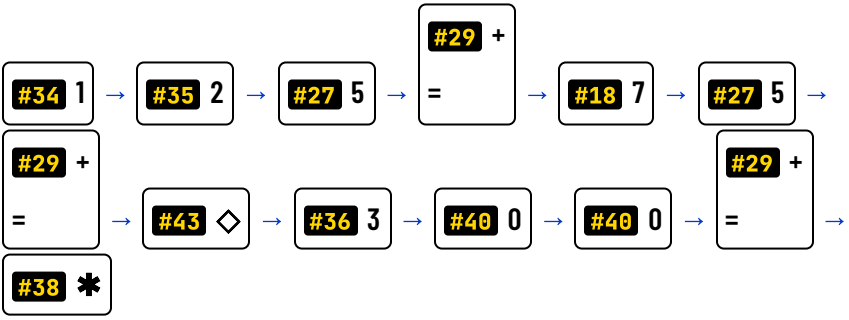
48. 7 (display 7.)



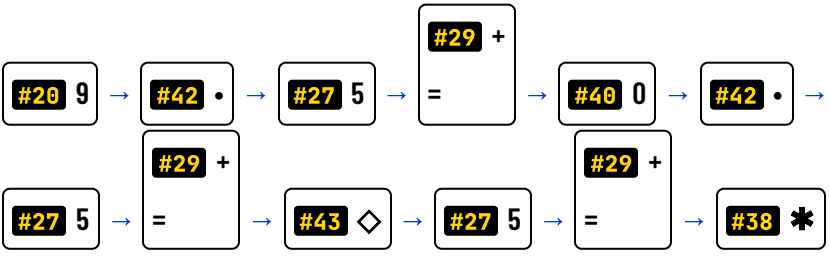
49. 50 (display 50.)



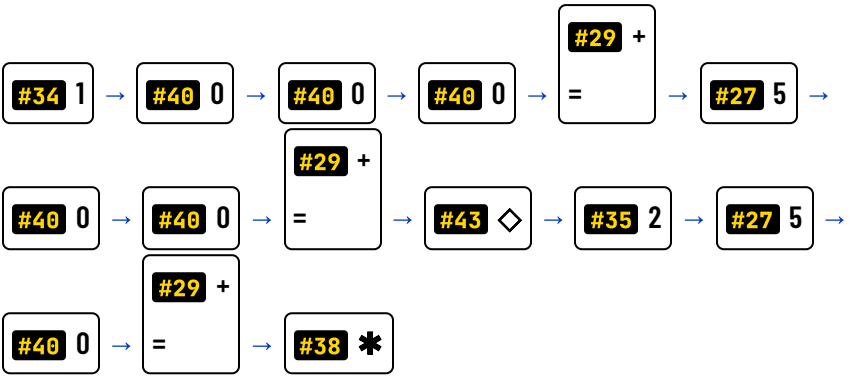
50. 500 (display 500.)



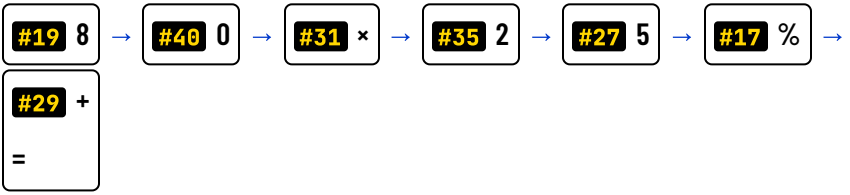
51. 15 (display 15.)



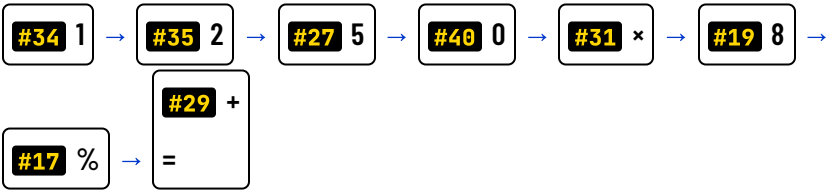
52. 1750 (display 1,750.)



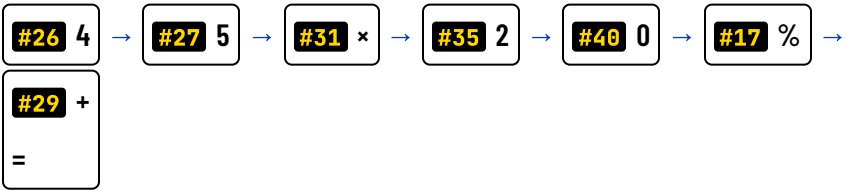
53. 100 (display 100.)



54. 1350 (display 1,350.)



55. 54 (display 54.)



56. 96 (display `96.`)

#28 6 → #26 4 → #31 × → #27 5 → #40 0 → #17 % → #29 + → =

57. 360 (display `360.`)

#36 3 → #35 2 → #40 0 → #31 × → #34 1 → #35 2 → #42 • → #27 5 → #17 % → #29 + → =

58. 159 (display `159.`)

#34 1 → #27 5 → #40 0 → #31 × → #28 6 → #17 % → #29 + → =

59. 60 (display `60.`)

#19 8 → #40 0 → #31 × → #35 2 → #27 5 → #17 % → #21 −

60. 1150 (display `1,150.`)

#34 1 → #35 2 → #27 5 → #40 0 → #31 × → #19 8 → #17 % → #21 −

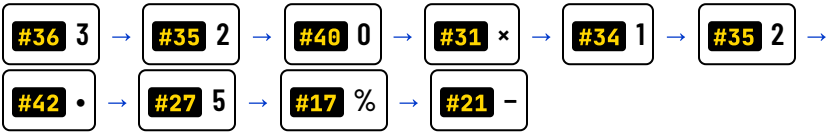
61. 36 (display `36.`)

#26 4 → #27 5 → #31 × → #35 2 → #40 0 → #17 % → #21 −

62. 32 (display `32.`)



63. 280 (display `280.`)



64. 141 (display `141.`)



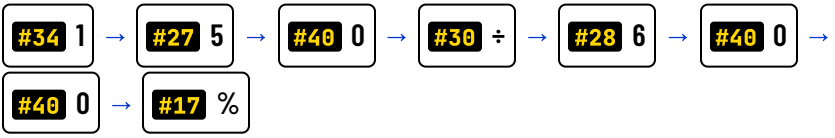
65. 25 (display `25.`)



66. 25 (display `25.`)



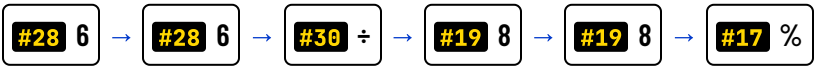
67. 25 (display `25.`)



68. 75 (display `75.`)



69. 75 (display `75.`)



70. 25 (display 25.)

#26 4 → #40 0 → #14 Δ% → #27 5 → #40 0 → #14 Δ%

71. 25 (display 25.)

#34 1 → #35 2 → #40 0 → #40 0 → #14 Δ% → #34 1 → #27 5 → #40 0 → #40 0 → #14 Δ%

72. 20 (display 20.)

#34 1 → #27 5 → #14 Δ% → #34 1 → #19 8 → #14 Δ%

73. 25 (display 25.)

#27 5 → #40 0 → #40 0 → #14 Δ% → #28 6 → #35 2 → #27 5 → #14 Δ%

74. 25 (display 25.)

#18 7 → #27 5 → #3 COST → #34 1 → #40 0 → #40 0 → #4 SELL

75. 40 (display 40.)

#34 1 → #35 2 → #40 0 → #3 COST → #35 2 → #40 0 → #40 0 → #4 SELL

76. 25 (display 25.)

#36 3 → #28 6 → #3 COST → #26 4 → #19 8 → #4 SELL

77. 20 (display 20.)

#26 4 → #40 0 → #40 0 → #3 COST → #27 5 → #40 0 → #40 0 → #4 SELL

78. 25 (display `25.`)

`#20` 9 → `#3` COST → `#34` 1 → `#35` 2 → `#4` SELL

79. 40 (display `40.`)

`#34` 1 → `#27` 5 → `#40` 0 → `#3` COST → `#35` 2 → `#27` 5 → `#40` 0 → `#4` SELL

80. 50 (display `50.`)

`#18` 7 → `#40` 0 → `#3` COST → `#34` 1 → `#26` 4 → `#40` 0 → `#4` SELL

81. 120 (display `120.`)

`#35` 2 → `#31` × → `#26` 4 → `#27` 5 → `#44` M+ → `#26` 4 → `#31` × → `#18` 7 → `#42` • → `#27` 5 → `#44` M+ → `#24` M✱

82. 41 (display `41.`)

`#34` 1 → `#40` 0 → `#31` × → `#36` 3 → `#42` • → `#35` 2 → `#44` M+ → `#28` 6 → `#31` × → `#34` 1 → `#42` • → `#27` 5 → `#44` M+ → `#24` M✱

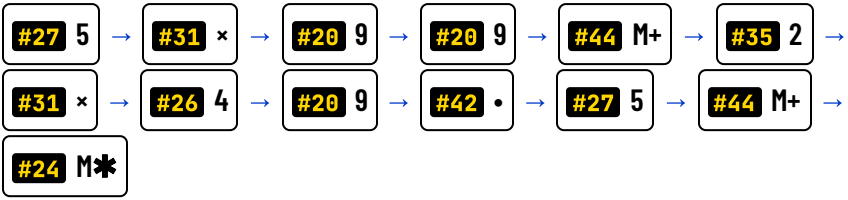
83. 420 (display `420.`)

`#34` 1 → `#35` 2 → `#31` × → `#35` 2 → `#27` 5 → `#44` M+ → `#36` 3 → `#31` × → `#26` 4 → `#40` 0 → `#44` M+ → `#24` M✱

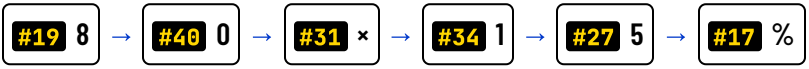
84. 113 (display `113.`)

`#18` 7 → `#31` × → `#18` 7 → `#44` M+ → `#19` 8 → `#31` × → `#19` 8 → `#44` M+ → `#24` M✱

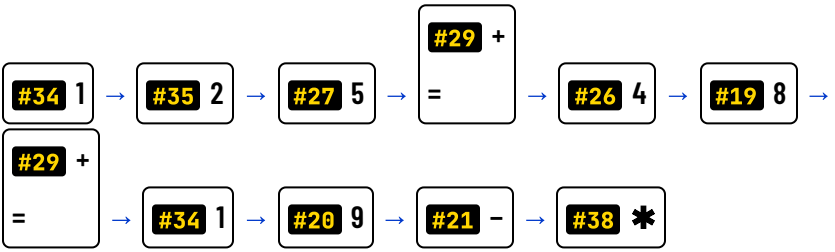
85. 594 (display `594.`)



86. 12 (display `12.`)



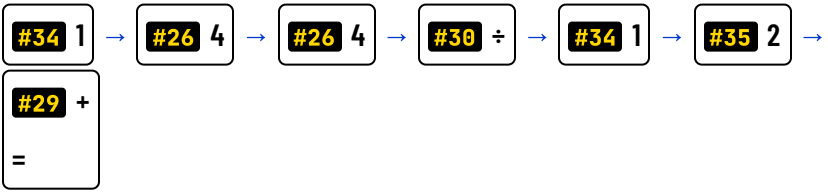
87. 154 (display `154.`)



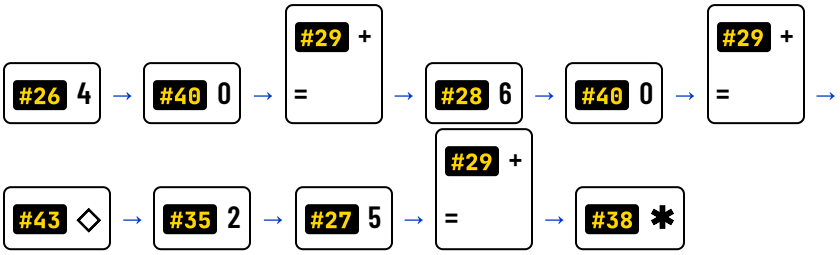
88. 180 (display `180.`)



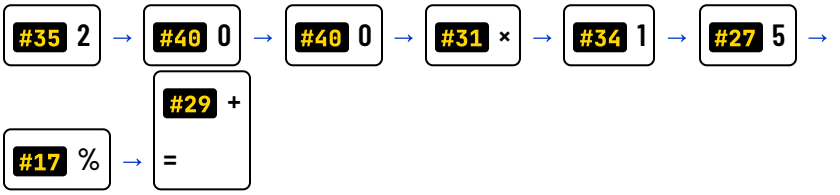
89. 12 (display `12.`)



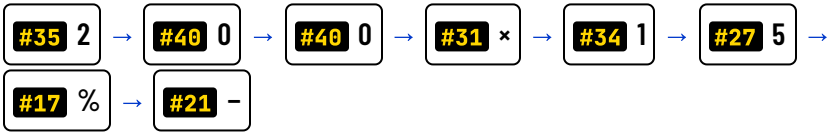
90. 125 (display `125.`)



91. 230 (display `230.`)



92. 170 (display `170.`)



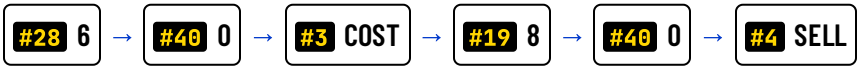
93. 25 (display `25.`)



94. 25 (display `25.`)



95. 25 (display `25.`)



96. 76 (display 76.)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #14
14. #15
15. #16
16. #17
17. #21
18. #22
19. #23
20. #24
21. #25
22. #29
23. #30
24. #31
25. #32
26. #33
27. #37
28. #38
29. #39
30. #41
31. #42
32. #43
33. #44

Appendix B — Function finder (A–Z)

Function	Label	Keys
# — non-add (print a reference number)	#	#12
% — percent (with + / − gives add-on / discount)	%	#17
+= — add the entry to the accumulator (adding-machine logic)	+=	#29
◇ — subtotal (print, keep running)	◇	#43
✱ — total (print and clear)	✱	#38
Backspace — delete the last digit typed	▶	#10
C — clear the entry	C	#33
CA — clear all (including accumulator)	CA	#9
Change sign (+/−) of the number being typed or the current result	+/−	#11
COST	COST	#3
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	•	#42
Digit 0	0	#40
Digit 1	1	#34
Digit 2	2	#35
Digit 3	3	#36
Digit 4	4	#26
Digit 5	5	#27
Digit 6	6	#28
Digit 7	7	#18
Digit 8	8	#19
Digit 9	9	#20
Division ÷	÷	#30
Double zero — types "00" in one press (fast entry of amounts)	00	#41
FEED — paper feed (blank line)	FEED	#8
GT — grand total	G✱	#37

IT — item count	IT	#2
M+ — add the value to independent memory M	M+	#44
M− — subtract the value from memory M	M−	#39
M◇ — memory subtotal (recall)	M◇	#32
M✱ — memory total (recall and clear)	M✱	#24
MAR — margin %	MAR	#5
MU — mark-up / mark-down	MUMD	#25
Multiplication ×	×	#31
Printed label only (no action in this app)	CONNECT	#13
PRT — print the display on the tape	PRT	#6
REPRINT — reprint the last calculation	REPRINT	#7
SELL	SELL	#4
Subtraction −	−	#21
Tax rate ▲ — increase the tax rate setting	▲	#15
Tax rate ▼ — decrease the tax rate setting	▼	#16
TAX+ — add tax at the set rate	TAX+	#23
TAX− — remove tax	TAX−	#22
TIME — print the time	TIME	#1
Δ% — percent change	Δ%	#14

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

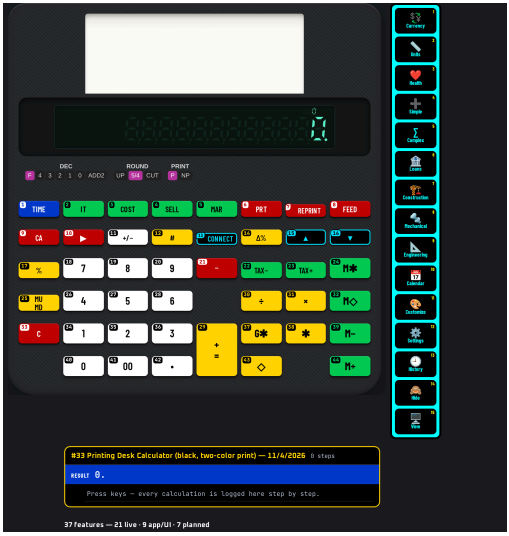
Two-colour printing desk calculator with on-screen paper tape (negative values in red). Adding-machine logic (+= / – accumulate, ✱ total, ◇ subtotal), item count, memory M+ M– M◇ M✱, grand total G✱, MU/MD mark-up, COST/SELL/MAR margin, TAX+/TAX– (rate: hold the ▲ / ▼ keys to change, default 8 %), Δ%, decimal (F 4 3 2 1 0 ADD2) and rounding (UP 5/4 CUT) slide switches, non-add/date print (#), REPRINT and FEED. TIME/DATE use the system clock; CONNECT (PC link) is a no-op.

Planned (not yet working)

- Currency conversion
- Tape export to PDF / CSV
- Tape search / filter
- Tape save & reload
- Speech read-out
- Colour-blind theme
- Custom tax presets

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 TIME	#2 IT	#3 COST
#4 SELL	#5 MAR	#6 PRT
#7 REPRINT	#8 FEED	#9 CA
#10 ▶	#11 +/−	#12 #
#13 CONNECT	#14 Δ%	#15 ▲
#16 ▼	#17 %	#18 7
#19 8	#20 9	#21 −
#22 TAX−	#23 TAX+	#24 M✱
#25 MUMD	#26 4	#27 5
#28 6	#29 +=	#30 ÷
#31 ×	#32 M◇	#33 C
#34 1	#35 2	#36 3
#37 G✱	#38 ✱	#39 M−
#40 0	#41 00	#42 •
#43 ◇	#44 M+	

U-45 CALCULATORS · Calculator #33 · cut out and keep